A'aeshah Alhakamy, Ph.D.

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Education

2017-2020 **Ph.D., in Computer Science.** *Purdue University,* Indianapolis, USA.
Thesis title: *Extraction and Integration of Physical Illumination in Dynamic Augmented Reality Environments.* https://doi.org/10.25394/PGS.12950315.v1

2015-2016 **M.Sc., in Computer Science.** *Purdue University,* Indianapolis, USA.

2001-2005 **B.Sc., in Computer Science.** *Taibah University,* Medina, Saudi Arabia.

Appointments

February 23, 2025 – Present **Associate Professor** of Computer Science, Faculty of Computers and Information Technology, University of Tabuk. Leading advanced research and teaching in computer graphics, computer vision, visualization, imaging, and extended reality. Promotion approved on December 22, 2024.

January 2025 – Present **Department Chair**, Department of Computer Science (Female Section), Faculty of Computers and Information Technology, University of Tabuk.

November 2024 – November 2025 **Deputy Supervisor**, Data Governance Unit, Vice Rectorate for Postgraduate Studies and Scientific Research, University of Tabuk.

September 2023 – September 2025 **Vice Chair**, Standing Executive Committee for Data Management and Governance, Data Governance Unit, Vice Rectorate for Postgraduate Studies and Scientific Research, University of Tabuk.

August 2023 – August 2025 **Data Management Lead**, Data Governance Unit, Vice Rectorate for Postgraduate Studies and Scientific Research, University of Tabuk.

June 2023 – December 2023 **Erasmus+ Liaison Officer**, Research and Consulting Institute, Vice Rectorate for Postgraduate Studies and Scientific Research, University of Tabuk.

May 2023 – Present **Member**, Computer Science Department Council, Faculty of Computers and Information Technology, University of Tabuk.

May 2021 – May 2023 **Member**, Information Technology Department Council, Faculty of Computers and Information Technology, University of Tabuk.

January 2023 – January 2025 **Researcher**, Artificial Intelligence and Sensing Technologies (AIST) Center, Vice Rectorate for Postgraduate Studies and Scientific Research, University of Tabuk.

December 2022 – December 2024 **Member**, Permanent Committee Supervising Social Responsibility Activities, Vice Rectorate for Postgraduate Studies and Scientific Research, University of Tabuk.

Appointments (continued)

April 2021 – December 2024 — **Assistant Professor** of Computer Science, Faculty of Computers and Information Technology, University of Tabuk; specializing in computer graphics, computer vision, visualization, imaging, and extended reality.

January 2023 – August 2023 — **Supervisor of Academic Affairs**, Department of Computer Science, Faculty of Computers and Information Technology, University of Tabuk.

January 2022 – January 2023 — **Saudi Federation for Cybersecurity, Programming and Drones (SAFCSP) Liaison Officer**, Research and Consulting Institute, Vice Rectorate for Postgraduate Studies and Scientific Research, University of Tabuk.

January 2021 – January 2023 — **Deputy Director**, Industrial Innovation and Robotics Center (IIRC), Vice Rectorate for Postgraduate Studies and Scientific Research, University of Tabuk.

Fall 2018 – Fall 2020 — **Graduate Research Assistant**, Human-Computer Interaction Program, Indiana University–Purdue University Indianapolis (IUPUI).

Fall 2017 — **Graduate Teaching Assistant**, Department of Computer Science, School of Science, Indiana University–Purdue University Indianapolis (IUPUI).

December 2009 – April 2021 — **Teaching Assistant** of Computer Science, Faculty of Computers and Information Technology, University of Tabuk.

2008 — **Computer Instructor**, Modern Girl Institute for Computer and Languages, and several high schools.

2007 — **Secretary**, Taibah Charity Association, Madinah.

Achievements

Awards

2026

- **Distinguished Community Partner Award**, Faculty Member Category, Deanship of Student Affairs, University of Tabuk. Post
- **Outstanding Scientific Research Award**, Web of Science–indexed journal, Deanship of Scientific Research, University of Tabuk; for the article `https://doi.org/10.1145/3652595`.

2025

- **Outstanding Scientific Research Award**, Web of Science–indexed journal, Deanship of Scientific Research, University of Tabuk; for the article `https://doi.org/10.1109/ACCESS.2025.3599337`.
- **Incentive Reward for Distinguished Researchers**, for high-citation scientific research, Deanship of Scientific Research, University of Tabuk.

2024

- **Excellence Award** (E-Learning, National Excellence Track), Teaching and Learning category, University of Tabuk; First Excellence Award cycle, 1445 AH/2024 AD. Post
- **Outstanding Faculty Member Award** in Student Activities, Deanship of Student Affairs, University of Tabuk. Post
- **Outstanding Scientific Research Award**, Web of Science–indexed journal, Deanship of Scientific Research, University of Tabuk; for the article `https://doi.org/10.1007/s11042-023-17598-7`.

Achievements (continued)

- **Outstanding Scientific Research Award**, Web of Science–indexed journal, Deanship of Scientific Research, University of Tabuk; for the article https://doi.org/10.3390/s23136024.

2023
- **Outstanding Scientific Research Award**, Web of Science–indexed journal, Deanship of Scientific Research, University of Tabuk; for the article https://doi.org/10.3390/sym15030699.
- **Outstanding Scientific Research Award**, Web of Science–indexed journal, Deanship of Scientific Research, University of Tabuk; for the article https://doi.org/10.3390/su15020911.
- **Incentive Reward for Distinguished Researchers**, for a publication coauthored with graduate students in a Web of Science–indexed journal, Deanship of Scientific Research, University of Tabuk; article http://dx.doi.org/10.14569/IJACSA.2022.0131246.

2022
- **President's Award for Outstanding Scientific Research**, Research and Innovation, Scientific and Engineering track, University of Tabuk. Post

2021
- **Outstanding Scientific Research Award**, Q1 Web of Science–indexed journal, Deanship of Scientific Research, University of Tabuk; for the article https://doi.org/10.1145/3386496.

2020
- **Travel Award**, to attend the CHI Conference; fully supported by Dr. Cafaro, NSF, and the IUPUI Data Lab.

2019
- **Outstanding Scientific Research Award**, for publishing as a Ph.D. student, Saudi Arabian Cultural Mission (SACM); article https://doi.org/10.1109/INFOCT.2019.8710982.

2018
- **Travel Award**, to attend the "Introduction to Academic Leadership" conference; funded by the Saudi Arabian Cultural Mission (SACM), including travel and accommodation.

2016
- **Master's Degree High-GPA Award**, Saudi Arabian Cultural Mission (SACM), for graduating with a high GPA.

2013
- **Scholarship**, full scholarship from the University of Tabuk and the Saudi Arabian Cultural Mission (SACM).

2005
- **Honors Degree**, graduated with Honors, B.Sc. in Computer Science, Taibah University.

Certifications

October 6–8, 2024
- **The 6 Critical Practices for Leading a Team: Supervisory Leadership Path**. Awarded by the Deanship of Quality and Academic Accreditation in cooperation with FranklinCovey.

June 5, 2023
- **Fundamentals of Developing and Building Policies and Legal Legislation**. Awarded by the National Data Management Office (NDMO).

July 24, 2021
- **Hadoop Platform and Application Framework**. Awarded by the University of California San Diego.

2018
- **Introduction to Academic Leadership**. Awarded by the Saudi Arabian Ministry of Education.

Achievements (continued)

Grants

August 2022 – May 2023 — **Deanship of Scientific Research, University of Tabuk**. Extended Reality (XR): Making immersive technologies in training and repair the key to unlocking Industry 4.0. Role: Principal Investigator (PI). Grant No.: S-1443-0002.

January 2019 – December 2020 — **Directorate for Computer & Information Science & Engineering (NSF/CISE)**. CHS:EAGER: Aiding Reasoning about Correlation and Causation. Role: Research Assistant (RA). URL: https://app.dimensions.ai/details/grant/grant.7706397. Grant No.: 1848898.

Mentoring (Research and Project Advising)

Master's Students

July 2022 – July 2023 — **Hibah Q. Alatawi, Nouf M. Albalawi, Ghada H. Shahata, and Khulud A. Aljohani**. Thesis: "Extended Reality (XR): Making Immersive Technologies in Training and Repair the Key to Unlocking Industry 4.0." Published article: "Augmented Reality-Assisted Deep Reinforcement Learning-Based Model toward Industrial Training and Maintenance for NanoDrop Spectrophotometer." http://doi.org/10.3390/s23136024.

May 2021 – May 2022 — **Shoroog Albalawi, Lama Alshahrani, Nouf Albalawi, and Reem Kilabi**. Thesis: "An Iris Recognition and Authentication System Using Image Analysis and Artificial Intelligence Techniques." Published article: "A Comprehensive Overview on Biometric Authentication Systems Using Artificial Intelligence Techniques." http://dx.doi.org/10.14569/IJACSA.2022.0130491.

Undergraduate Students

May 2025 – Present — **Awatif Al-Huwaiti, Yara Al-Ghamdi, Manal Al-Subaie, Fatima Al-Balharith, Ahoud Al-Huwaiti, Maram Al-Fahmani, Lina Al-Anzi**. Capstone project: "Nakhwa."

Lubna Al-Nafie, Rawan Al-Otaibi, Ghadeer Al-Shahrani, Rimas Al-Enezi, Thuraya Al-Madani, Badour Al-Rashidi, Sama Al-Sarhani. Capstone project: "VeriNews."

August 2024 – May 2025 — **Maram Al-Zahrani, Raghad Al-Zahrani, Raghad Al-Anzi, Aryaf Al-Mutairi, Lama Hamlan**. Capstone project: "Smart Coach."

Nour Al Omrani, Rimas Al Zahrani, Al Maha Al Balawi, Dhikra Al Atwi, Al Jouri Al Hindi. Capstone project: "Visual Text."

January 2024 – May 2024 — **Nour Mubarak, Bayader Al-Issa, Maryam Majrashi, Shawq Al-Fihmani, Nada Al-Adisani, Aseel Al-Atwi**. Capstone project: "YUSR: A Mobile Application for Umrah and Hajj in Makkah."

August 2022 – February 2023 — **Shahad Alsufyani, Alanoud Albalawi, Maha Al-Shammari, Madawi Al-Anzi, Albatul Altoum, Maha Alenazi, and Alia Alatwi**. Capstone project: "Road Breakers."

Heba Al-Balawi, Wejdan Al-Atwi, Batool Abdullah, Wadha Al-Hwaiti, Bayan Al-Atwi, Samah Alhawiti, Maryam Alatwi, and Razan Alatwi. Capstone project: "UT Repository."

August 2021 – May 2022 — **Renad Alsharafaa, Manar Albalawi, Awatef Alharthi, Njoud Alhazmi, Amal Aamri, and Razan Asiri**. Capstone project: "Stay."

Mentoring (Research and Project Advising) (continued)

- **Rana Alharbi, Hadeel Aljuhani, Ahad Aljohani, Abeer Albalawi, Ghada Albushi, and Rahaf Al-Atwi**. Capstone project: "Student Club."

Gifted Students

July 2024 – December 2024

- Supervised scientific projects for students participating in the National Science Olympiad – Creativity 2025, General Administration of Education in the Tabuk Region (Educational Affairs – Capacity Development Department – Gifted Students Department).
- Supervised and supported scientific projects for students participating in the Mawhiba competition for the National Science Olympiad – Creativity 2025, Tabuk International Schools.

University and Community Services

Keynote Talks and Workshops

September 10, 2025 — **Flowchart Design Workshop** — Deanship of Student Affairs, University of Tabuk, Saudi Arabia; BNG Program.

September 8, 2025 — **Design Sketching Workshop** — Deanship of Student Affairs, University of Tabuk, Saudi Arabia; BNG Program.

September 19, 2024 — **Technical Podcast** — Deanship of Student Affairs, University of Tabuk, Saudi Arabia; recorded an episode on "Digital Fraud."

January 28 – February 1, 2024 — **Professional Center** — Deanship of Student Affairs with the Research and Consulting Institute, University of Tabuk, Saudi Arabia; delivered a 40-hour training program titled "Cybersecurity," in cooperation with the King Abdulaziz Charitable Society and King Abdulaziz Model Schools.

February 9, 2023 — **Deanship of Scientific Research**, University of Tabuk, Saudi Arabia; scientific seminar within the Research Hour Initiative titled "Extended Reality (XR) toward Building Immersive Solutions: The Key to Unlocking Industry 6.0" for the funded project (Grant No.: S-1443-0002).

November 8, 2021 — **Faculty of Computers and Information Technology**, University of Tabuk, Saudi Arabia; seminar titled "Essential Steps to Create Your Own Website."

September 16, 2021 — **Digital Giving Webinar (Ataa)**; seminar titled "External Scholarship and Self-Education to Keep Pace with Digital Transformation."

September 8, 2021 — **Faculty of Computers and Information Technology**, University of Tabuk, Saudi Arabia; seminar titled "Extraction and Integration of Physical Illumination in Dynamic Augmented Reality Environments."

Reviewer, Technical Committee, and Session Chair

March 8–11, 2026 — **ACM TEI 2026: 20th International Conference on Tangible, Embedded, and Embodied Interaction**, Chicago, Illinois, USA.

September 11, 2025 — **Information Processing & Management** (Elsevier); reviewer (two manuscripts).

University and Community Services (continued)

April 13–14, 2025 — **IEEE 2025 4th International Conference on Computing and Information Technology (ICCIT)**, Grand Millennium, University of Tabuk, Tabuk, Saudi Arabia; Session Chair.

November 11–13, 2024 — **IEEE 2024 11th International Conference on Control, Mechatronics and Automation (ICCMA 2024)**, Brunel University, London, UK.

June 12–14, 2024 — **ACM International Conference on Interactive Media Experiences (IMX 2024)**, Stockholm, Sweden.

November 1–3, 2023 — **IEEE 2023 11th International Conference on Control, Mechatronics and Automation (ICCMA 2023)**, Campus Grimstad, University of Agder, Norway.

September 13–14, 2023 — **IEEE 2023 3rd International Conference on Computing and Information Technology (ICCIT)**, Grand Millennium, University of Tabuk, Tabuk, Saudi Arabia; Session Chair and participant.

December 2022 – 2023 — **Multimedia Tools and Applications** (Springer Nature); reviewer (16 manuscripts).

April 2022 – 2023 — **International Journal of Human–Computer Interaction** (Taylor & Francis), London, UK; reviewer (two manuscripts).

February 13–16, 2022 — **ACM TEI 2022: 16th International Conference on Tangible, Embedded, and Embodied Interaction**, KAIST, Daejeon, South Korea.

November 30 – December 2, 2021 — **IEEE 2021 2nd International Conference on Computing and Information Technology (ICCIT)**, University of Tabuk, Saudi Arabia.

Ambassador

February 9, 2023 — **Future Ambassadors Program, Ministry of Communications and Information Technology**, Riyadh, Saudi Arabia; delivered a workshop titled "Extended Reality (XR) toward Building Immersive Solutions: The Key to Unlocking Industry 4.0."

December 12–20, 2022 — **Future Ambassadors Program, Ministry of Communications and Information Technology**, Riyadh, Saudi Arabia; trained selected students from multiple universities across the Kingdom on the "Digital 101" curriculum, delivering six two-hour workshops.

January 15 – December 20, 2022 — **Future Ambassadors Program, Ministry of Communications and Information Technology**, Riyadh, Saudi Arabia; Stanford Center for Professional Development Program: Digital 101; attended six two-hour workshops.

Event Mentor and Judge

April 29–30, 2026 — **X-thon 4 Road Safety Hackathon** — Deanship of Student Affairs, University of Tabuk, Saudi Arabia; Judge, Technical Committee.

October 9, 2025 — **BNG: To Build Innovative Minds Challenge** — Deanship of Student Affairs, University of Tabuk, Saudi Arabia; Judge.

July 24, 2025 — **Jadah30, Social Development Bank**, Tabuk, Saudi Arabia; judge at the JadahThon Hackathon focused on innovative digital solutions for e-commerce and entrepreneurship.

University and Community Services (continued)

February 1, 2025 — **Tabuk International Schools**, Tabuk, Saudi Arabia; supervised and supported students in the "Mawhiba" Competition for Scientific Creativity 2025.

General Administration of Education in the Tabuk Region (Educational Affairs – Capacity Development Department – Gifted Students Section), Saudi Arabia; provided guidance and support for students' scientific projects participating in the National Science Olympiad – Creativity 2025.

November 6, 2024 — **BNG: To Build Innovative Minds Challenge** — Deanship of Student Affairs, University of Tabuk, Saudi Arabia; Judge.

October 5, 2024 — **NASA International Space Apps Challenge** — Deanship of Student Affairs, University of Tabuk, Saudi Arabia; Judge.

February 4, 2024 — **|2|-Thon** — Deanship of Student Affairs, University of Tabuk, Saudi Arabia; Judge, Scientific Patents Track.

January 18 and 28, 2024 — **Princess Seetah Award** — Deanship of Student Affairs, University of Tabuk, Saudi Arabia; substitute judge, Sustainable Solutions Track.

January 27, 2024 — **2nd Global Game Jam** — Google Developer Student Clubs (GDSC), Deanship of Student Affairs, University of Tabuk, Saudi Arabia; Judge.

October 7, 2023 — **NASA International Space Apps Challenge** — Deanship of Student Affairs, University of Tabuk, Saudi Arabia; Judge.

February 2–5, 2023 — **Xthon-2** — Deanship of Student Affairs, University of Tabuk, Saudi Arabia; Judge, Asynchronous Learning Track.

September 13–15, 2022 — **Global Game Jam** — Google Developer Student Clubs (GDSC), Deanship of Student Affairs, University of Tabuk, Saudi Arabia; mentor and judge.

January 3–7, 2021 — **Xthon-1** — Deanship of Student Affairs, University of Tabuk, Saudi Arabia; mentor, Sustainable Environment Applications Challenge.

Presenter

November 30 – December 2, 2021 — **IEEE 2021 2nd International Conference on Computing and Information Technology (ICCIT)**, University of Tabuk, Saudi Arabia; event presenter.

Participant

December 19–20, 2023 — **Digital Governance Forum**, Riyadh, Saudi Arabia; explored cooperation opportunities for the Data Management Office at the University of Tabuk.

December 1, 2022 – January 1, 2023 — **Digital Innovation Award, Ministry of Communications and Information Technology**, hybrid (Riyadh and online); participated with the research project "SARA for Diabetic Children" and attended workshops and a month-long training program.

September 13–15, 2022 — **Research Partnership Day**, Grand Millennium, Vice Rectorate for Postgraduate Studies and Scientific Research, University of Tabuk; participated in the session on protecting natural reserves.

Global Summit on Artificial Intelligence, Riyadh, Saudi Arabia; attended by special invitation.

University and Community Services (continued)

April 19, 2022 — **International Education Conference and Exhibition**, Riyadh, Saudi Arabia; attended conference activities and workshops; represented the university at its booth.

February 1–3, 2022 — **LEAP 2022**, Riyadh, Saudi Arabia; explored cooperation opportunities for the Industrial Innovation and Robotics Center.

October 22, 2021 — **The Knowledge Society Series in E-Learning Workshop**, Deanship of E-Learning and Distance Education, University of Tabuk, Tabuk, Saudi Arabia; participated in discussions.

Teaching

Spring 2026 – Second Semester 1447

CIT 1305 — **Database Systems**, 6 credit hours (lecture), B.Sc. in IT.

CIS 601 — **Fundamentals of Data Science**, 3 credit hours (lecture), M.Sc. in Data Science .

Fall 2025 – First Semester 1447

CSC 1301 — **Software Engineering**, 6 credit hours (lecture), B.Sc. in CS/IT.

CIS 601 — **Fundamentals of Data Science**, 3 credit hours (lecture), M.Sc. in AI.

Summer 2025 – Summer Semester 1446

CSC 390 — **Field Training for CS Students**, 3 credit hours, B.Sc. in CS.

CIT 390 — **Field Training for IT Students**, 3 credit hours, B.Sc. in IT.

Spring 2025 – Second Semester 1446

CIT 1305 — **Database Systems**, 6 credit hours (lecture), B.Sc. in IT.

CSC 490 — **Graduation Project 1**, 2 credit hours, B.Sc. in CS/IT.

CSC 491 — **Graduation Project 2**, 6 credit hours, B.Sc. in CS/IT.

CIS 611 — **Big Data Analytics**, 3 credit hours (lecture), M.Sc. in Data Science .

Fall 2024 – First Semester 1446

CSC 420 — **Theory of Computation**, 6 credit hours (lecture), B.Sc. in CS.

CSC 1201 — **Object-Oriented Programming**, 6 credit hours (lecture), B.Sc. in CS.

CSC 490 — **Graduation Project 1**, 1 credit hour, B.Sc. in CS.

CSC 491 — **Graduation Project 2**, 3 credit hours, B.Sc. in CS.

CSC 606 — **Computer Vision**, 3 credit hours (lecture), M.Sc. in AI.

Spring 2024 – Second Semester 1445

CSC 420 — **Theory of Computation**, 3 credit hours (lecture), B.Sc. in CS.

CSC 1203 — **Advanced Programming MethoData Science** , 9 credit hours (lecture), B.Sc. in CS.

CSC 1101 — **Introduction to Computing**, 2 credit hours (lecture), B.Sc. in CS.

CSC 490 — **Graduation Project 1**, 1 credit hour, B.Sc. in CS.

CIT 616 — **Introduction to Cybersecurity and Digital Forensics**, 3 credit hours (lecture), Master of Forensic Accounting.

Fall 2023 – First Semester 1445

CSC 420 — **Theory of Computation**, 9 credit hours (lecture), B.Sc. in CS.

CIT 453 — **Multimedia Systems**, 3 credit hours (lecture), B.Sc. in IT.

CSC 1102 — **Problem-Solving and Programming Concepts**, 2 credit hours (lecture), B.Sc. in CS.

Teaching (continued)

CIS 611 **Big Data Analytics**, 3 credit hours (lecture), M.Sc. in Data Science .

Spring 2023 – Third Trimester 1444

CIT 0453 **Multimedia Systems**, 15 credit hours (lecture), B.Sc. in IT.

CSC 0624 **Research Project 3**, 3 credit hours, M.Sc. in AI.

Winter 2022 – Second Trimester 1444

CSC 0452 **Computer Graphics**, 5 credit hours (lecture), B.Sc. in CS.

CSC 0492 **Selected Topics in Computer Science**, 5 credit hours, B.Sc. in CS.

CSC 0491 **Graduation Project 2**, 3 credit hours, B.Sc. in CS.

CIT 0491 **Graduation Project 2**, 3 credit hours, B.Sc. in IT.

CIT 0603 **Database Security**, 6 credit hours (5 lecture + 1 lab), Master of Information Security.

CSC 0624 **Research Project 2**, 3 credit hours, M.Sc. in AI.

Fall 2022 – First Trimester 1444

CIT 0341 **IT Project Management**, 8 credit hours (lecture), B.Sc. in IT.

CSC 0490 **Graduation Project 1**, 1 credit hour, B.Sc. in CS.

CIT 0490 **Graduation Project 1**, 1 credit hour, B.Sc. in IT.

CSC 1101 **Introduction to Computing**, 5 credit hours (lecture), B.Sc. in CS.

CIS 0611 **Big Data Analytics**, 4 credit hours (lecture), M.Sc. in Data Science .

CIT 0616 **Introduction to Cybersecurity and Digital Forensics**, 4 credit hours (lecture), Master of Forensic Accounting.

CSC 0624 **Research Project 1**, 3 credit hours, M.Sc. in AI.

Spring 2022 – Second Semester 1443

CSC 490 **Graduation Project 1**, 3 credit hours, B.Sc. in CS.

Graduation Project 1, 6 credit hours, B.Sc. in CS.

CSC 101 **Computer Programming 1**, 3 credit hours (lecture), B.Sc. in CS.

CIS 601 **Fundamentals of Data Science**, 3 credit hours (lecture), M.Sc. in AI.

CSC 624 **Research Project 2**, 3 credit hours, M.Sc. in AI.

Fall 2021 – First Semester 1443

CSC 490 **Graduation Project 1**, 2 credit hours, B.Sc. in CS.

CSC 102 **Computer Programming 2**, 6 credit hours (lecture), B.Sc. in CS.

CIS 611 **Big Data Analytics**, 3 credit hours (lecture), M.Sc. in Data Science .

CIS 605 **Programming for Data Science**, 3 credit hours (lecture), M.Sc. in Data Science .

CSC 624 **Research Project 1**, 3 credit hours, M.Sc. in AI.

Spring 2021 – Second Semester 1442

CSC 102 **Computer Programming 2**, 12 credit hours (lecture) and 2 credit hours (lab), B.Sc. in CS.

Spring 2013 – Second Semester 1433

CSC 420 **Theory of Computation**, 3 credit hours (lecture), B.Sc. in CS.

CSC 210 **Computer Organization with Assembly**, 2 credit hours (lab), B.Sc. in CS.

CSC 491 **Graduation Project**, 2 credit hours (lab), B.Sc. in CS.

Fall 2012 – First Semester 1433

CSC 210 **Computer Organization with Assembly**, 6 credit hours (lab), B.Sc. in CS.

Teaching (continued)

CSC 220 **Data Structures and Algorithms**, 4 credit hours (lab), B.Sc. in CS.

Spring 2011 – Second Semester 1432

CSC 101 **Computer Programming 1**, 2 credit hours (lab), B.Sc. in CS.

CSC 102 **Computer Programming 2**, 6 credit hours (lab), B.Sc. in CS.

Fall 2011 – First Semester 1432

DM 241 **Introduction to Computers**, 2 credit hours (lecture) and 2 credit hours (lab), B.Sc. in CS.

CSC 112 **Programming Languages**, 8 credit hours (lab), B.Sc. in CS.

CSC 109 **Computer Ethics**, 2 credit hours (lecture), B.Sc. in CS.

Spring 2010 – Second Semester 1431

CSC 102 **Computer Programming 2**, 2 credit hours (lab), B.Sc. in CS.

Fall 2010 – First Semester 1431

CE 5 224 **Data Structures**, 3 credit hours (lecture), Community College, Associate Diploma.

Research Publications

Journal Articles

1. Alhakamy, A. (2026). Managing uncertainty and information dynamics with graphics-enhanced togaf architecture in higher education. *Entropy*, *28*(3). doi:10.3390/e28030361
2. Alharbi, L. A., Alnahas, J., Javed, A., Alhakamy, A., Nawaz, M., Malik, H., … Alfaifi, F. (2026). Deeprawnet: Empowering deepfake audio detection through dynamic enhancements. *PeerJ Computer Science*, *12*, e3670. doi:10.7717/peerj-cs.3670
3. Alhakamy, A. (2025a). Advancing sdg5: Machine learning and statistical graphics for women's empowerment and gender equity. *Sustainability*, *17*(21). doi:10.3390/su17219706
4. Friedman, A., Tazike, M., Gokpinar Shelton, E., Patel, N., Alhakamy, A., & Cafaro, F. (2025). "i'm a fish!": Exploring children's engagement with human–data interactions in museums. *Applied Sciences*, *15*(21). doi:10.3390/app152111304
5. Alhakamy, A. (2025b). Intersecting realms: Examining the convergence of vision and ai in extended reality graphics. *IEEE Access*, 1–1. doi:10.1109/ACCESS.2025.3599337
6. Albalawi, S., Alamri, L., Atut, J., Albalawi, S., Haddaddi, R., & Alhakamy, A. (2025). Mahya: Facial recognition-based pilgrim identification system for enhanced health monitoring and assistance. *International Journal of Advanced Computer Science and Applications*, *16*(3). doi:10.14569/IJACSA.2025.0160391
7. Alhakamy, A. (2024a). Extended reality (xr) toward building immersive solutions: The key to unlocking industry 4.0. *ACM Comput. Surv.*, *56*(9). doi:10.1145/3652595
8. Alhakamy, A. (2023a). Digital vaccination passport and administrative burden of covid-19 official application based on twitter(x) discourse: Public interest vs democracy. *Multimedia Tools and Applications*, *23*(7). doi:10.1007/s11042-023-17598-7
9. Alatawi, H., Albalawi, N., Shahata, G., Aljohani, K., Alhakamy, A., & Tuceryan, M. (2023). Augmented reality-assisted deep reinforcement learning-based model towards industrial training and maintenance for nanodrop spectrophotometer. *Sensors*, *23*(13). doi:10.3390/s23136024
10. Alhakamy, A. (2023b). Fathoming the mandela effect: Deploying reinforcement learning to untangle the multiverse. *Symmetry*, *15*(3). doi:10.3390/sym15030699

11. Alhakamy, A., Alhowaity, A., Alatawi, A. A., & Alsaadi, H. (2023). Are used cars more sustainable? price prediction based on linear regression. *Sustainability*, *15*(2). doi:10.3390/su15020911

12. Albalawi, S., Alshahrani, L., Albalawi, N., Kilabi, R., & Alhakamy, A. (2022). A comprehensive overview on biometric authentication systems using artificial intelligence techniques. *International Journal of Advanced Computer Science and Applications*, *13*(4). doi:10.14569/IJACSA.2022.0130491

13. Alharbi, R., Alageel, N., Alsayil, M., Alharbi, R., & Alhakamy, A. (2022). Prediction of oil production through linear regression model and big data tools. *International Journal of Advanced Computer Science and Applications*, *13*(12). doi:10.14569/IJACSA.2022.0131246

14. Alomani, S. M., Alhawiti, N. I., & Alhakamy, A. (2022). Prediction of quality of water according to a random forest classifier. *International Journal of Advanced Computer Science and Applications*, *13*(6). doi:10.14569/IJACSA.2022.01306105

15. Alhakamy, A., & Tuceryan, M. (2020a). Real-time illumination and visual coherence for photorealistic augmented/mixed reality. *ACM Comput. Surv.*, *53*(3). doi:10.1145/3386496

16. Trajkova, M., Alhakamy, A., Cafaro, F., Vedak, S., Mallappa, R., & Kankara, S. R. (2020). Exploring casual covid-19 data visualizations on twitter: Topics and challenges. *Informatics*, *7*(3). doi:10.3390/informatics7030035

Conference Proceedings

1. Friedman, A., Tazike, M., Shelton, E. G., Patel, N. A., Alhakamy, A., & Cafaro, F. (2025). Puppeteers and forecasters: How children engage. In *Proceedings of the 2025 acm symposium on spatial user interaction*. doi:10.1145/3694907.3765950

2. Alhakamy, A. (2024b). "xr is a game": Exploring people's reasons for using, limiting, and abandoning extended reality. In *2024 ieee international conference on real-time computing and robotics (rcar)* (pp. 271–278). doi:10.1109/RCAR61438.2024.10670927

3. Alessa, N., Majdua, A., Alshehri, S., Alhawiti, M., Aljohani, R., & Alhakamy, A. (2023). Banking in terms of deposit prediction based on machine learning and big data analytics. In *2023 11th international conference on control, mechatronics and automation (iccma)* (pp. 69–74). doi:10.1109/ICCMA59762.2023.10374595

4. Alshammari, A., Alshammari, R., Altalak, M., Alshammari, K., & Alhakamy, A. (2022). Credit-card fraud detection system using big data analytics. In *2022 international conference on electrical, computer, communications and mechatronics engineering (iceccme)* (pp. 1–7). doi:10.1109/ICECCME55909.2022.9987791

5. Alhakamy, A., Trajkova, M., & Cafaro, F. (2021). Show me how you interact, i will tell you what you think: Exploring the effect of the interaction style on users' sensemaking about correlation and causation in data. In *Designing interactive systems conference 2021* (pp. 564–575). doi:10.1145/3461778.3462083

6. Alhakamy, A., Cafaro, F., Trajkova, M., Kankara, S., Mallappa, R., & Veda, S. (2020). Design strategies and optimizations for human-data interaction systems in museums. In *2020 ieee 20th international conference on advanced learning technologies (icalt)* (pp. 246–248). doi:10.1109/ICALT49669.2020.00081

7. Trajkova, M., Alhakamy, A., Cafaro, F., Mallappa, R., & Kankara, S. R. (2020). Move your body: Engaging museum visitors with human-data interaction. In *Proceedings of the 2020 chi conference on human factors in computing systems* (pp. 1–13). doi:10.1145/3313831.3376186

8. Alhakamy, A., & Tuceryan, M. (2019a). An empirical evaluation of the performance of real-time illumination approaches: Realistic scenes in augmented reality. In L. T. De Paolis & P. Bourdot (Eds.), *Augmented reality, virtual reality, and computer graphics* (pp. 179–195). doi:10.1007/978-3-030-25999-0_16

9 Alhakamy, A., & Tuceryan, M. (2019b). Polarization-based illumination detection for coherent augmented reality scene rendering in dynamic environments. In M. Gavrilova, J. Chang, N. M. Thalmann, E. Hitzer, & H. Ishikawa (Eds.), *Advances in computer graphics* (pp. 3–14). doi:10.1007/978-3-030-22514-8_1

10 Alhakamy, A., & Tuceryan, M. (2019c). Ar360: Dynamic illumination for augmented reality with real-time interaction. In *2019 ieee 2nd international conference on information and computer technologies (icict)* (pp. 170–174). doi:10.1109/INFOCT.2019.8710982

11 Alhakamy, A., & Tuceryan, M. (2019d). Cubemap360: Interactive global illumination for augmented reality in dynamic environment. In *2019 southeastcon* (pp. 1–8). doi:10.1109/SoutheastCon42311.2019.9020588

12 Alsaiari, A., Rustagi, R., Alhakamy, A., Thomas, M. M., & Forbes, A. G. (2019). Image denoising using a generative adversarial network. In *2019 ieee 2nd international conference on information and computer technologies (icict)* (pp. 126–132). doi:10.1109/INFOCT.2019.8710893

13 Cafaro, F., Trajkova, M., & Alhakamy, A. (2019). Designing embodied interactions for informal learning: Two open research challenges. In *Proceedings of the 8th acm international symposium on pervasive displays*. doi:10.1145/3321335.3329688

Books and Chapters

1 Alhakamy, A., & Tuceryan, M. (2020b). *Physical environment reconstruction beyond light polarization for coherent augmented reality scene on mobile devices* (M. L. Gavrilova, C. J. K. Tan, J. Chang, & N. M. Thalmann, Eds.). doi:10.1007/978-3-662-61983-4_2

Recommendations

Available on LinkedIn and upon Request